

Practice Problems : Eng Prac in Maths and Comput (MA 122)
B.Tech. Ist Year, 2025
Course Instructor: MKR+SK

1. Consider the following initial value problem:

$$\frac{dy}{dt} = -2ty^2, \quad y(t=0) = 1$$

Write a MATLAB code to compute $y(10)$ using Euler method with $\Delta t = 0.2$.

2. Consider the following system of equations:

$$\begin{aligned}\frac{dy}{dt} &= z \\ \frac{dz}{dt} + 5tz + (y+7)\sin t &= 0 \\ y(t=0) &= 5, \quad z(t=0) = 1.4\end{aligned}$$

Write a MATLAB code to compute the solutions of the above system using the Euler method at $t = 10$ with a time step of $\Delta t = 0.5$.
